

Advanced (Habanero)

- Q1.** What is the slope of the line $y = 2x + 1$?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q2.** A line passes through points (1,2) and (3,4). What is its slope?
(A) 0
(B) 1
(C) 2
(D) $\frac{1}{2}$
(E) $\frac{2}{3}$
- Q3.** What is the slope of a horizontal line?
(A) Positive
(B) Negative
(C) Zero
(D) Undefined
(E) None of the above
- Q4.** The equation $y = -3x + 2$ represents a line with what kind of slope?
(A) Positive
(B) Negative
(C) Zero
(D) Undefined
(E) None of the above
- Q5.** A line has a slope of -2 . Which statement is true about this line?
(A) It rises 2 units for every 1 unit it runs to the right
(B) It falls 2 units for every 1 unit it runs to the right
(C) It rises 1 unit for every 2 units it runs to the right
(D) It falls 1 unit for every 2 units it runs to the right
(E) None of the above
- Q6.** What is the slope of the line $x = 4$?
(A) 0
(B) 1
(C) 2
(D) 3
(E) Undefined
- Q7.** A water tank is being filled at a rate of 2 gallons per minute. What is the slope of the graph of the amount of water in the tank versus time?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q8.** A bike travels 20 miles in 2 hours. What is the slope of the graph of the distance traveled versus time?
(A) 5
(B) 10
(C) 15
(D) 20
(E) 25
- Q9.** A line has a slope of -2 . Which statement is true about

Open Ended Questions (FRQ)

- Q9.** Find the slope of the line passing through points (2,3) and (4,5). Show your work and explain your reasoning.
- Q10.** A car travels 120 miles in 4 hours. Find the slope of the graph of the distance traveled versus time. Explain what this slope represents in the context of the problem.

Chef's Tips

- Emphasize the concept of slope as rate of change
- Use graphs, tables, and equations to represent linear relationships
- Have students interpret slope in context